

# Smart Edge for Cloud-Ready Data

*Real-time data processing at the edge, built for privacy and efficiency*

*Connecting a device is no longer enough to call it 'smart'. The real challenge lies in real-time data refining: operating on milliwatts while ensuring total privacy at the edge*

*Des données orientées Cloud directement depuis les systèmes embarqués Dati orientati al Cloud direttamente dai dispositivi integrati*

**Below are some critical challenges — and opportunities — where embedded systems design directly impacts the effectiveness of real-time data engineering at the edge**

## **DATA COLLECTION (SOURCE)**

Capturing raw signals at the point of origin

## **DATA INGESTION (PIPELINE)**

Efficiently moving bits into local memory

## **DATA PROCESSING (REFINING)**

Transforming noisy waveforms into structured, meaningful features

## **DATA QUALITY (VALIDATION)**

Ensuring integrity before any data leaves the device

## **DATA STORAGE (PERSISTENCE)**

Local buffering and temporary storage for fault-tolerant streaming

## **DATA ANALYTICS (INTELLIGENCE)**

Lightweight computation at the milliwatt level, near the sensor

## **DATA TRANSFORMATION (SYNTHESIS)**

Compressing and enriching data to high-value payloads

## **DATA INTEGRATION (TRANSPORT)**

Sending only refined, relevant data to the cloud

## **DATA GOVERNANCE (LIFECYCLE)**

Managing versions, metadata, and signal health on the edge

# Cloud Integration Map

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| EDGE FUNCTION | AWS STACK                | GCP STACK                  |
|---------------|--------------------------|----------------------------|
| Ingestion     | IoT Core / Kinesis       | IoT Core / Pub-Sub         |
| Processing    | Lambda / Glue            | Cloud Functions / Dataflow |
| Analytics     | Redshift/Timestream / S3 | BigQuery / Cloud Storage   |
| Governance    | IoT Device Management    | Fleet Management           |

Refining the Cloud pipeline at the Edge: Intelligence at the point of origin

# Sensors & Real-Time Data Processing

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## HIGH-SPEED DATA STREAMING

Low-latency data pipelines targeting deterministic sub-millisecond latency — an ideal design goal for real-time, cloud-native processing at the edge

## HANDLING RAW BINARY SIGNALS

On-device data validation and filtering to remove noise and outliers before cloud ingestion

## EDGE COMPUTING ARCHITECTURE

Edge-native architectures enabling local synthesis and significant bandwidth reduction before data reaches the cloud

# Data Efficiency & Payload Optimization

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## **SMART DOWNSAMPLING**

Reducing data volume while preserving informational entropy

## **LOCAL AGGREGATION**

Transforming continuous raw signals into discrete, high-level events

# Hardware-Level Security & Data Privacy

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## **DATA ANONYMIZATION**

On-chip hashing and encryption to enforce privacy by design

## **ZERO RAW-DATA CLOUD EXPOSURE**

Raw signals remain on-device, with only refined and contextualized insights transmitted upstream

# TinyML: On-Device Data Intelligence

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## FEATURE ENGINEERING

Extracting meaningful features directly on constrained devices

## ANOMALY DETECTION

Local pattern recognition to detect abnormal behavior in real time

## QUANTIZATION

Reducing model size and compute requirements while preserving accuracy



# Towards Intelligent Edge

*Improving cloud data processes starts with better data at the source*

**Which of these challenges is most critical for your project?**

Quel point est critique pour votre projet ?

Quale di questi punti è critico per il tuo progetto ?